ANDREW NEIL

AI ENGINEER | AGENTIC AI & PYTHON BACKEND

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PROFESSIONAL SUMMARY

Computer Engineering student (2027) focused on agentic AI, LLM-powered automation and Python backend engineering. Builds tool-using systems that translate natural-language goals into structured actions, with hands-on experience in Django workflows, browser and desktop automation, computer vision, and test-driven reliability. Interested in AI Engineer, Agentic AI Engineer and Python Backend Engineer roles.

TECHNICAL SKILLS

Programming: Python, JavaScript, TypeScript, SQL, C/C++
AI / Agent Engineering: LLM APIs (Gemini, OpenAI, Claude), local LLMs (Ollama), tool calling, agent orchestration, structured outputs, LangChain, LangGraph
Automation / Vision: Playwright, Selenium, PyAutoGUI, OpenCV, YOLOv8, OCR, ANPR
Backend / Data: Django, FastAPI, REST APIs, relational databases, Pandas, NumPy, Parquet, pytest
Platforms / Tools: Git, GitHub, Linux CLI, Flutter, React, Vite, Tailwind CSS, ESP32, RFID / NFC, OmniRoute

EXPERIENCE

Backend Engineer Intern | 4iStudio | *Jul 2026 - Oct 2026*
- Contributed Python and Django backend workflows for a college ERP, supporting student records and academic administration across departments, teachers, classes and students.
- Worked on marks entry, attendance, student-data management, marksheets and certificate workflows.

Freelance Python Developer | Upwork | *2023 - 2025*
- Delivered Python automation and browser-scraping solutions, including an appointment-availability monitor with automated email alerts through client SMTP.
- Built a command-line LLM workflow that generated multiple response variants and reused the selected response as context; completed three fixed-price contracts and earned two five-star reviews.

SELECTED PROJECTS

OSCAR - Vision-Powered Desktop Automation Agent | *Personal project* | GitHub
- Built a Windows-first, local-first agentic automation system with a Python orchestration backend and Flutter desktop interface; converts natural-language tasks into structured actions across browser, desktop UI and terminal tools.
- Integrated model-provider routing, Playwright, PyAutoGUI, screenshot/OCR observations and record/replay; added bounded retries, outcome checks and regression tests (243 passing at the latest recorded run).

TradeAssist - Quantitative Research & Backtesting | *Personal project*
- Developed an Indian-equities research and risk-analysis workflow with technical indicators, modeled brokerage/slippage, stop-loss rules and 13 walk-forward folds.
- Best recorded IRFC backtest (SMA 20/50, 5% stop-loss; Jan 2023-Sep 2026): +24.17% cumulative return, 6.75% CAGR, -11.03% maximum drawdown across 86 trades; historical simulation, not live performance.

FinVerse 2.0 - Agentic B2B Payment Gateway Prototype | *Final-year project in development*
- Exploring machine-to-machine agentic commerce through a B2B payment workflow, AI copilot, React dashboard, Django/Python backend and ESP32-S3/NFC authentication terminal.
- Researches blockchain-assisted auditability and hardware-backed authentication; prototype is under development and is not a production payment gateway.

Dynamic Toll Charging System | *Competition-winning project*
- Combined YOLOv8 vehicle detection, OpenCV, OCR/ANPR and RFID identification to demonstrate vehicle matching and simulated FASTag-style toll charging; co-authored an IJIRT paper in 2026.

KYC Smart Card | *IoT and document-security project* | GitHub
- Designed a Django and Flutter document platform with ESP32 RFID smartcard verification, Bluetooth LE mobile authentication, JWT and document encryption; associated with two IJIRT publications.

EDUCATION

B.E. in Computer Engineering | Xavier Institute of Engineering (XIE), Mumbai | Expected 2027
CGPA: 7.48/10 (through Semester VI) | Coursework: Data Structures, DBMS, Operating Systems, Computer Networks, AI/ML, IoT

SELECTED PUBLICATIONS

- “Dynamic Toll Charging System,” IJIRT, Vol. 12, Issue 8, 2026.
- “KYC Smart Card” (dual-mode RFID / Bluetooth authentication), IJIRT, Vol. 12, Issue 1, June 2025.
- “KYC Smart Card” (RFID document repository), IJIRT, Vol. 11, Issue 7, December 2024.